

1-8B Llama vs. Mamba2 vs. GLA2 vs. GDN5 || Tasks: Babilong (left), NIAH-S (right)

8B	1.0T(4096)	Llama - original - Ir0.003	-	51.3%		75.5%	70.0%	59.7%	42.5%	46.1%	40.1%	32.2%	25.3%	30.1%	36.4%	30.7%	25.9%	58.1%	53.1%	49.6%	49.4%	82.4%	82.5%	71.6%	64.1%			100.0%	99.0%	100.0%	0.0%	0.0%	0.0%	0.0%	100.0%	100.0%	100.0%	100.0%	0.0%	0.0%	0.0%	100.0%	98.0%	93.2%	90.6%	0.0%	0.0%	0.0%	
8B	1.0T(4096)	Llama - original - Ir0.002	-	48.5%		69.4%	68.1%	59.4%	37.6%	41.1%	33.0%	31.8%	23.5%	28.5%	35.2%	31.2%	22.8%	56.7%	48.5%	48.7%	49.6%	81.0%	73.9%	66.0%	64.5%			100.0%	99.8%	97.4%	0.0%	0.0%	0.0%	0.0%	100.0%	100.0%	100.0%	100.0%	0.0%	0.0%	0.0%	100.0%	99.8%	83.6%	70.0%	0.0%	0.0%	0.0%	
8B	1.0T(4096)	Llama - Canon -ABCD - Ir0.003	-	53.7%		84.7%	79.0%	72.9%	49.0%	55.2%	41.2%	36.1%	26.1%	36.5%	39.1%	31.8%	27.5%	58.7%	49.9%	42.5%	43.5%	86.9%	82.5%	68.9%	62.3%			100.0%	100.0%	100.0%	5.0%	0.0%	0.0%	0.0%	100.0%	100.0%	100.0%	100.0%	0.0%	0.0%	0.0%	100.0%	100.0%	99.6%	99.8%	0.0%	0.0%	0.0%	
8B	1.0T(4096)	Llama - Canon -ABCD - Ir0.002	-	52.1%		82.5%	78.8%	67.6%	53.0%	44.6%	37.2%	33.0%	28.1%	25.7%	29.7%	28.7%	23.9%	53.5%	49.5%	44.5%	52.8%	82.3%	82.0%	74.2%	71.1%			100.0%	100.0%	100.0%	0.2%	0.0%	0.0%	0.0%	100.0%	100.0%	100.0%	100.0%	0.0%	0.0%	0.0%	100.0%	100.0%	100.0%	98.4%	0.0%	0.0%	0.0%	
8B	1.0T(4096)	Mamba2 - conv - Ir0.003	-	37.6%		59.8%	55.5%	44.7%	28.5%	31.0%	24.4%	19.6%	16.8%	24.7%	25.3%	20.4%	16.8%	55.9%	35.9%	31.8%	27.1%	67.6%	62.1%	55.5%	48.9%			100.0%	100.0%	100.0%	100.0%	100.0%	99.8%	99.6%	100.0%	100.0%	87.0%	85.4%	31.8%	0.0%	0.0%	100.0%	98.0%	64.4%	59.4%	18.4%	0.0%	0.0%	
8B	1.0T(4096)	Mamba2 - conv - Ir0.002	-	36.8%		55.1%	51.6%	45.2%	31.2%	36.6%	29.4%	20.9%	16.3%	26.9%	25.3%	23.6%	19.4%	53.4%	34.0%	29.5%	28.1%	43.4%	54.3%	55.3%	55.7%			100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	99.2%	98.4%	41.6%	4.6%	0.0%	100.0%	95.4%	66.4%	60.6%	18.8%	2.4%	0.0%	
8B	1.0T(4096)	Mamba2 - Canon -AbCD - Ir0.003	-	40.1%		65.8%	49.5%	38.2%	32.5%	37.5%	30.3%	23.7%	21.3%	34.8%	32.9%	26.1%	22.9%	47.5%	38.2%	33.1%	35.9%	66.7%	62.0%	54.5%	48.7%			100.0%	100.0%	100.0%	100.0%	100.0%	97.4%	32.6%	100.0%	100.0%	90.4%	84.8%	18.4%	5.4%	0.0%	99.6%	89.6%	55.2%	53.0%	7.6%	3.8%	0.0%	
8B	1.0T(4096)	Mamba2 - Canon -AbCD - Ir0.002	-	40.6%		55.4%	49.6%	42.5%	35.3%	36.6%	31.4%	28.5%	21.0%	27.5%	31.2%	28.9%	22.9%	57.5%	46.5%	41.0%	37.6%	60.3%	58.0%	52.1%	48.8%			100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	97.0%	93.8%	4.4%	0.0%	0.0%	100.0%	98.4%	61.6%	64.0%	3.8%	0.0%	0.0%	
8B	1.0T(4096)	GLA5 - conv - Ir0.003	-	42.4%		67.3%	65.0%	50.3%	38.6%	39.7%	29.5%	21.1%	18.0%	35.9%	30.6%	22.9%	21.4%	56.9%	41.9%	32.3%	27.5%	60.4%	65.5%	62.3%	53.3%			100.0%	100.0%	98.8%	94.6%	87.6%	64.4%	45.6%	100.0%	100.0%	66.2%	63.8%	72.4%	61.2%	59.4%	99.8%	99.6%	74.4%	71.0%	63.4%	56.8%	46.2%	
8B	1.0T(4096)	GLA5 - conv - Ir0.002	-	43.5%		67.5%	56.6%	47.4%	33.7%	45.0%	32.7%	28.6%	19.8%	34.6%	30.2%	27.0%	21.3%	59.3%	50.6%	38.8%	32.9%	68.3%	62.3%	57.6%	55.8%			100.0%	100.0%	100.0%	99.8%	98.2%	93.4%	83.2%	100.0%	100.0%	60.0%	61.2%	70.8%	52.6%	63.4%	100.0%	98.4%	77.8%	77.0%	61.6%	60.6%	52.4%	
8B	1.0T(4096)	GLA5 - Canon -AbCD - Ir0.003	-	39.6%		62.4%	47.1%	33.5%	24.1%	40.4%	33.8%	22.8%	17.3%	36.7%	33.5%	26.6%	22.7%	56.5%	46.9%	36.7%	26.0%	68.5%	55.4%	52.9%	47.4%			100.0%	100.0%	100.0%	99.2%	97.2%	90.4%	78.2%	100.0%	100.0%	99.0%	98.4%	91.8%	89.4%	86.6%	99.8%	94.4%	87.0%	83.4%	74.0%	71.2%	64.2%	
8B	1.0T(4096)	GLA5 - Canon -AbCD - Ir0.002	-	40.9%		55.4%	48.3%	44.5%	34.5%	38.4%	34.1%	25.5%	17.5%	26.8%	29.0%	27.6%	22.5%	57.6%	45.8%	36.6%	33.1%	67.6%	63.6%	57.1%	53.2%			100.0%	100.0%	99.8%	100.0%	99.4%	99.4%	98.8%	100.0%	100.0%	96.0%	99.4%	78.4%	79.6%	78.0%	99.6%	99.2%	87.0%	87.0%	66.4%	63.0%	51.2%	
8B	1.0T(4096)	GDN2 - conv - Ir0.003	-	41.6%		57.1%	52.1%	42.9%	31.9%	37.6%	28.2%	22.8%	19.7%	30.0%	27.3%	23.5%	18.6%	63.2%	50.9%	44.3%	38.1%	66.1%	63.1%	59.9%	54.8%			100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	94.8%	95.6%	95.4%	84.4%	88.2%	100.0%	98.6%	93.8%	93.6%	77.0%	61.4%	58.6%
8B	1.0T(4096)	GDN2 - conv - Ir0.002	-	39.8%		46.6%	47.5%	40.1%	30.5%	41.9%	33.2%	24.9%	19.5%	28.7%	30.4%	27.9%	21.8%	61.7%	45.7%	35.4%	30.2%	62.3%	58.2%	55.3%	53.4%			100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	87.4%	90.0%	69.2%	63.0%	66.6%	100.0%	99.0%	91.6%	88.4%	41.0%	40.8%	36.8%
8B	1.0T(4096)	GDN2 - Canon -AbCD - Ir0.003	-	43.6%		64.9%	55.4%	47.1%	43.0%	40.3%	35.8%	30.6%	22.9%	32.6%	31.4%	25.1%	22.5%	55.2%	46.0%	38.8%	32.1%	71.6%	63.4%	56.6%	55.6%			100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	87.6%	100.0%	99.4%	95.2%	92.8%	100.0%	99.4%	86.4%	88.0%	76.8%	72.6%	61.0%
8B	1.0T(4096)	GDN2 - Canon -AbCD - Ir0.002	-	43.0%		71.6%	51.4%	40.6%	31.1%	42.0%	33.7%	28.1%	21.8%	31.9%	31.3%	28.2%	25.9%	58.7%	45.0%	37.1%	31.6%	71.3%	60.7%	60.9%	57.5%			100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	99.8%	100.0%	99.8%	99.0%	98.8%	100.0%	99.6%	89.8%	86.4%	87.4%	74.2%	66.0%

3B 1.0T(4096) Llama - original - Ir0.003	38.9%	49.6%	62.8%	55.2%	15.8%	37.8%	31.7%	25.3%	6.1%	27.2%	27.4%	22.8%	4.2%	52.8%	56.7%	47.1%	30.8%	70.5%	63.8%	69.4%	20.3%	100.0%	100.0%	99.8%	0.0%	0.0%	0.0%	0.0%	100.0%	100.0%	100.0%	0.0%	0.0%	0.0%	0.0%	99.8%	98.8%	98.2%	0.0%	0.0%	0.0%	0.0%		
3B 1.0T(4096) Llama - original - Ir0.002	40.9%	57.6%	54.1%	61.5%	28.0%	37.7%	31.1%	26.4%	7.6%	24.9%	31.2%	24.0%	3.2%	44.1%	53.2%	46.1%	27.9%	79.8%	74.0%	69.2%	36.4%	100.0%	100.0%	100.0%	0.0%	0.0%	0.0%	0.0%	100.0%	100.0%	100.0%	0.0%	0.0%	0.0%	0.0%	100.0%	99.6%	96.8%	0.0%	0.0%	0.0%	0.0%		
3B 1.0T(4096) Llama - Canon -ABCD - Ir0.003	43.9%	64.5%	55.1%	45.2%	23.3%	43.6%	30.3%	26.8%	23.3%	38.1%	32.1%	28.2%	25.3%	52.9%	41.1%	42.1%	51.1%	74.5%	65.7%	60.3%	53.6%	100.0%	100.0%	100.0%	100.0%	0.0%	0.0%	0.0%	0.0%	100.0%	100.0%	100.0%	100.0%	0.8%	0.0%	0.0%	100.0%	100.0%	100.0%	96.8%	0.0%	0.0%	0.0%	0.0%
3B 1.0T(4096) Llama - Canon -ABCD - Ir0.002	49.0%	67.3%	68.2%	60.5%	51.7%	42.2%	34.4%	28.9%	26.6%	35.7%	34.1%	31.3%	22.3%	52.0%	43.4%	42.2%	50.4%	80.4%	72.5%	69.6%	66.3%	100.0%	100.0%	99.6%	99.4%	2.0%	0.0%	0.0%	0.0%	100.0%	100.0%	100.0%	100.0%	2.4%	0.0%	0.0%	100.0%	97.4%	96.6%	99.2%	0.0%	0.0%	0.0%	0.0%
3B 1.0T(12288) Mamba2 - conv - Ir0.005	36.3%	46.9%	50.5%	39.2%	35.0%	33.6%	26.4%	18.4%	16.3%	29.5%	26.9%	23.2%	18.7%	46.3%	34.1%	32.8%	29.1%	58.5%	57.7%	52.7%	49.8%	100.0%	100.0%	100.0%	100.0%	100.0%	99.6%	99.2%	100.0%	99.8%	98.4%	98.6%	93.2%	80.2%	75.8%	99.8%	87.0%	58.4%	50.6%	35.6%	36.0%	38.0%		
3B 1.0T(12288) Mamba2 - conv - Ir0.003	35.3%	43.8%	46.6%	38.9%	29.7%	34.6%	23.9%	20.7%	16.1%	27.1%	24.6%	21.7%	17.5%	48.0%	33.6%	33.4%	32.4%	55.8%	58.6%	52.1%	46.5%	100.0%	100.0%	99.8%	99.8%	99.8%	100.0%	100.0%	100.0%	100.0%	100.0%	99.2%	98.4%	89.8%	85.6%	84.2%	100.0%	81.4%	74.8%	66.2%	46.2%	49.0%	36.2%	
3B 1.0T(12288) Mamba2 - conv - Ir0.002	36.6%	50.0%	46.6%	40.6%	32.3%	36.3%	25.2%	18.6%	14.7%	33.4%	25.9%	21.1%	16.7%	52.6%	34.3%	31.6%	28.9%	65.3%	55.9%	53.3%	49.1%	100.0%	100.0%	98.8%	97.6%	95.8%	93.4%	91.4%	99.8%	100.0%	99.8%	97.8%	90.4%	83.2%	82.2%	99.8%	84.2%	33.2%	41.4%	10.2%	11.0%	15.2%		
3B 1.0T(12288) Mamba2 - Canon -AbCD - Ir0.005	36.7%	47.5%	44.1%	38.6%	34.5%	31.7%	23.2%	19.6%	16.1%	28.2%	28.7%	24.0%	18.0%	54.9%	38.9%	32.0%	30.4%	67.6%	58.5%	52.5%	45.3%	100.0%	100.0%	99.4%	100.0%	99.0%	99.6%	99.4%	100.0%	100.0%	97.6%	94.0%	67.6%	66.4%	56.2%	100.0%	98.2%	78.6%	75.4%	57.0%	58.8%	54.6%		
3B 1.0T(12288) Mamba2 - Canon -AbCD - Ir0.003	33.6%	50.8%	31.0%	28.0%	30.6%	33.6%	23.8%	17.8%	16.4%	19.9%	22.5%	21.3%	21.6%	49.4%	39.5%	35.0%	31.1%	59.2%	51.8%	43.7%	45.3%	100.0%	100.0%	99.8%	100.0%	99.8%	100.0%	99.8%	100.0%	99.8%	97.2%	79.4%	64.2%	72.6%	99.8%	87.8%	55.6%	60.4%	45.2%	45.8%	55.2%			
3B 1.0T(12288) Mamba2 - Canon -ABCD - Ir0.002	35.5%	47.4%	48.1%	41.6%	35.5%	30.6%	26.2%	18.8%	15.3%	25.0%	26.0%	22.1%	19.1%	49.7%	38.8%	32.0%	30.6%	59.3%	49.6%	48.0%	45.8%	100.0%	99.8%	99.4%	99.0%	98.6%	97.0%	95.8%	100.0%	100.0%	96.6%	92.2%	82.6%	73.0%	71.2%	99.8%	65.8%	65.8%	62.6%	51.4%	46.2%	43.4%		
3B 1.0T(12288) GLA5 - conv - Ir0.005	39.9%	62.4%	49.5%	42.2%	37.5%	34.4%	27.8%	22.9%	20.1%	30.7%	28.3%	24.7%	23.3%	53.6%	43.2%	40.3%	31.1%	62.9%	58.2%	55.2%	49.3%	100.0%	99.8%	98.4%	97.4%	92.2%	89.2%	81.2%	100.0%	100.0%	90.6%	87.6%	78.2%	59.6%	59.0%	99.8%	88.4%	59.2%	47.2%	46.6%	45.6%	30.0%		
3B 1.0T(12288) GLA5 - conv - Ir0.003	40.3%	55.7%	53.3%	46.5%	34.3%	38.9%	28.2%	23.8%	19.6%	29.8%	28.2%	24.1%	22.4%	56.1%	43.8%	33.7%	31.9%	65.9%	62.7%	55.7%	50.3%	100.0%	100.0%	98.6%	97.2%	89.4%	82.8%	74.8%	100.0%	100.0%	86.6%	90.2%	88.4%	88.6%	80.0%	100.0%	94.2%	84.4%	79.2%	64.8%	56.6%	52.2%		
3B 1.0T(12288) GLA5 - conv - Ir0.002	37.5%	52.0%	46.9%	42.9%	35.8%	38.3%	27.6%	23.6%	18.5%	29.4%	29.6%	24.8%	21.2%	53.0%	39.1%	33.6%	29.3%	51.1%	53.7%	49.2%	51.0%	100.0%	100.0%	97.6%	96.2%	91.8%	88.6%	83.0%	100.0%	100.0%	96.0%	92.0%	55.2%	51.6%	47.0%	100.0%	92.8%	68.6%	59.0%	30.0%	30.0%	23.2%		
3B 1.0T(12288) GLA5 - Canon -AbCD - Ir0.005	37.9%	54.5%	38.9%	33.2%	29.2%	33.6%	30.3%	25.4%	19.8%	27.9%	27.5%	26.8%	17.5%	59.2%	43.6%	35.5%	35.0%	62.2%	53.8%	51.0%	53.1%	100.0%	100.0%	99.0%	99.0%	98.8%	98.8%	97.0%	100.0%	100.0%	99.2%	97.4%	92.4%	93.0%	88.2%	100.0%	99.6%	91.4%	90.0%	82.2%	72.4%	62.4%		
3B 1.0T(12288) GLA5 - Canon -ABCD - Ir0.003	40.5%	49.6%	50.7%	42.6%	34.8%	36.8%	31.0%	23.1%	19.3%	32.4%	30.9%	29.9%	23.5%	50.1%	46.2%	40.3%	37.9%	68.2%	59.7%	54.0%	48.8%	100.0%	99.8%	98.2%	99.0%	89.0%	84.2%	70.6%	100.0%	100.0%	98.4%	98.8%	74.6%	79.4%	76.4%	100.0%	96.6%	59.4%	61.4%	26.8%	29.8%	19.4%		
3B 1.0T(12288) GLA5 - Canon -AbCD - Ir0.002	40.0%	48.8%	49.4%	45.4%	39.2%	35.7%	30.8%	24.8%	19.8%	26.4%	31.4%	25.7%	17.4%	51.0%	44.6%	35.6%	31.9%	62.1%	65.2%	58.6%	55.5%	100.0%	100.0%	98.2%	98.0%	94.4%	91.8%	84.0%	100.0%	100.0%	99.4%	99.2%	92.6%	91.2%	95.0%	100.0%	99.2%	92.8%	81.4%	69.2%	65.8%	56.4%		
3B 1.0T(12288) GDN2 - conv - Ir0.005	39.3%	48.3%	49.0%	44.5%	40.8%	37.9%	29.6%	25.6%	22.5%	27.5%	28.0%	25.6%	24.4%	57.3%	42.6%	37.5%	34.2%	53.6%	52.0%	49.9%	55.3%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	97.2%	92.4%	91.4%	100.0%	99.6%	95.6%	89.6%	82.2%	85.0%	82.4%			
3B 1.0T(12288) GDN2 - conv - Ir0.003	37.2%	45.5%	44.1%	39.6%	35.2%	34.2%	30.0%	22.0%	17.5%	23.1%	24.8%	25.1%	21.3%	49.5%	43.4%	39.0%	32.9%	51.8%	54.4%	53.8%	56.2%	100.0%	100.0%	100.0%	100.0%	100.0%	99.8%	100.0%	100.0%	100.0%	98.0%	98.0%	90.0%	87.0%	82.6%	100.0%	98.4%	88.2%	87.6%	64.0%	65.4%	63.2%		
3B 1.0T(12288) GDN2 - conv - Ir0.002	37.3%	51.5%	43.9%	40.2%	33.5%	34.5%	27.0%	22.5%	18.5%	22.6%	25.9%	23.7%	20.3%	54.8%	40.4%	33.2%	29.4%	65.8%	57.6%	50.8%	49.7%	100.0%	100.0%	100.0%	100.0%	99.8%	99.8%	99.6%	100.0%	100.0%	99.4%	99.2%	69.4%	67.8%	60.6%	100.0%	94.2%	71.0%	58.4%	9.4%	37.2%	56.8%		
3B 1.0T(12288) GDN2 - Canon -AbCD - Ir0.005	40.9%	62.3%	51.9%	39.8%	36.8%	40.7%	27.8%	22.1%	20.0%	27.2%	29.2%	26.3%	21.2%	56.6%	41.9%	37.6%	35.2%	70.7%	57.5%	55.9%	58.2%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	97.6%	97.0%	95.2%	85.0%	67.2%	100.0%	98.0%	94.0%	92.2%	86.6%	90.8%	65.6%	
3B 1.0T(12288) GDN2 - Canon -AbCD - Ir0.003	39.5%	43.5%	50.2%	43.9%	39.8%	37.4%	32.8%	25.2%	20.4%	24.0%	26.6%	25.1%	21.8%	57.1%	38.5%	33.8%	34.2%	61.3%	63.2%	55.9%	54.7%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	94.8%	94.2%	86.8%	82.0%	74.4%	100.0%	96.0%	78.2%	81.2%	79.8%	68.6%	74.6%	
3B 1.0T(12288) GDN2 - Canon -AbCD - Ir0.002	38.7%	54.6%	40.0%	39.4%	30.8%	40.5%	33.4%	26.4%	23.4%	25.2%	25.6%	24.8%	23.2%	55.7%	40.2%	33.9%	35.5%	62.8%	58.9%	53.7%	46.2%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	98.8%	99.2%	98.0%	87.4%	91.6%	100.0%	95.4%	90.8%	77.0%	61.8%	65.2%	64.2%	

1.3B	1.0T(4096)	Llama - original - Ir0.003	-34.7%	51.2%	64.9%	45.7%	8.3%	34.7%	27.1%	23.0%	3.8%	26.4%	26.0%	18.7%	4.1%	55.0%	45.8%	37.9%	11.4%	64.8%	70.7%	65.6%	8.9%	100.0%	99.6%	82.4%	0.0%	0.0%	0.0%	0.0%	100.0%	100.0%	99.8%	0.0%	0.0%	0.0%	0.0%	99.8%	99.8%	85.0%	0.0%	0.0%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																																
1.3B	1.0T(4096)	Llama - original - Ir0.002	-34.1%	55.9%	61.4%	43.5%	8.9%	36.3%	29.5%	24.2%	4.1%	29.6%	32.6%	20.5%	2.7%	50.6%	45.0%	37.2%	12.2%	49.5%	61.8%	66.3%	10.9%	100.0%	91.0%	73.0%	0.0%	0.0%	0.0%	0.0%	100.0%	100.0%	98.2%	0.0%	0.0%	0.0%	0.0%	99.8%	88.4%	57.2%	0.0%	0.0%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																																
1.3B	1.0T(4096)	Llama - Canon-ABCD - Ir0.003	-43.2%	53.5%	66.3%	56.1%	44.2%	39.3%	30.3%	24.6%	21.6%	29.0%	26.8%	27.7%	24.3%	45.3%	40.9%	45.6%	40.7%	59.6%	61.1%	67.1%	59.8%	100.0%	100.0%	99.6%	100.0%	57.8%	32.8%	23.6%	100.0%	100.0%	100.0%	99.8%	53.6%	0.0%	0.0%	0.0%	100.0%	99.4%	99.8%	97.6%	45.0%	2.4%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(4096)	Llama - Canon-ABCD - Ir0.002	-41.6%	57.3%	55.9%	46.0%	21.9%	33.2%	28.9%	20.7%	17.3%	27.5%	27.5%	26.4%	19.4%	49.5%	38.5%	41.4%	41.2%	73.6%	69.1%	69.6%	67.3%	100.0%	100.0%	100.0%	100.0%	12.4%	9.6%	0.0%	100.0%	99.6%	100.0%	100.0%	83.6%	0.0%	0.2%	99.8%	99.2%	99.6%	97.6%	28.8%	1.8%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																															
1.3B	1.0T(16384)	Mamba2 - conv - Ir0.005	-34.8%	48.3%	44.9%	35.7%	31.1%	29.0%	22.4%	20.4%	16.7%	23.6%	20.1%	15.2%	18.3%	54.4%	33.5%	31.1%	33.0%	62.3%	55.1%	52.4%	49.0%	100.0%	100.0%	99.8%	99.6%	99.2%	98.0%	96.6%	100.0%	99.8%	96.4%	91.4%	73.4%	79.0%	74.2%	99.6%	74.6%	41.0%	20.0%	22.8%	21.8%	5.4%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	Mamba2 - conv - Ir0.003	-34.8%	42.7%	40.8%	37.4%	39.2%	31.3%	22.8%	16.6%	15.2%	23.3%	22.4%	17.9%	12.1%	45.8%	31.7%	31.2%	28.5%	61.1%	60.5%	59.6%	56.5%	99.8%	100.0%	99.2%	98.6%	92.0%	84.0%	68.2%	100.0%	99.4%	95.2%	93.0%	80.8%	73.6%	72.6%	79.8%	65.8%	46.0%	43.2%	37.2%	33.6%	31.8%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	Mamba2 - conv - Ir0.002	-35.3%	48.6%	42.5%	33.4%	30.9%	30.4%	21.1%	19.8%	16.5%	21.5%	21.3%	19.3%	13.6%	55.3%	34.8%	33.0%	30.7%	61.3%	60.8%	59.0%	51.1%	100.0%	99.6%	96.8%	97.2%	90.2%	85.8%	70.2%	100.0%	100.0%	97.0%	94.2%	79.2%	67.4%	69.0%	98.6%	63.4%	21.8%	10.2%	11.4%	3.2%	11.8%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	Mamba2 - Canon-AbCD - Ir0.005	-33.8%	43.6%	31.6%	29.0%	27.6%	33.2%	23.3%	18.5%	13.8%	21.9%	25.4%	17.1%	14.0%	51.9%	40.1%	36.1%	31.6%	52.2%	55.4%	56.9%	53.4%	100.0%	100.0%	99.4%	99.2%	95.8%	95.6%	87.2%	100.0%	99.6%	95.4%	85.4%	70.6%	59.8%	58.6%	98.4%	81.0%	31.0%	36.2%	5.4%	2.8%	2.8%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	Mamba2 - Canon-AbCD - Ir0.003	-36.6%	46.7%	37.3%	33.0%	34.4%	34.5%	22.1%	21.9%	19.5%	24.5%	23.5%	18.0%	17.3%	48.8%	41.8%	39.1%	36.3%	66.5%	56.8%	55.5%	53.6%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	99.6%	100.0%	99.0%	88.4%	77.8%	61.8%	61.2%	61.8%	99.2%	43.8%	12.2%	2.8%	2.8%	1.8%	2.4%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	Mamba2 - Canon-ABCD - Ir0.002	-35.4%	38.8%	41.4%	35.1%	32.6%	30.5%	26.2%	20.4%	21.2%	20.4%	21.5%	21.9%	17.8%	48.0%	40.5%	36.3%	28.0%	58.7%	61.0%	56.3%	51.4%	100.0%	100.0%	100.0%	99.8%	99.4%	100.0%	99.0%	100.0%	99.6%	78.0%	74.2%	62.0%	61.8%	60.2%	99.4%	65.2%	50.8%	43.0%	28.8%	31.0%	31.0%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	GLA5 - conv - Ir0.005	-34.8%	49.9%	38.5%	34.3%	30.6%	35.8%	22.0%	18.6%	18.1%	21.5%	24.8%	18.5%	11.2%	51.9%	33.3%	32.0%	30.6%	56.4%	55.9%	55.6%	56.7%	100.0%	100.0%	95.8%	93.6%	79.4%	73.6%	63.2%	100.0%	100.0%	96.6%	94.4%	37.8%	37.4%	40.2%	99.0%	92.6%	46.2%	36.0%	17.4%	26.0%	26.4%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	GLA5 - conv - Ir0.003	-36.6%	54.2%	46.4%	38.3%	36.9%	36.0%	22.1%	15.5%	13.7%	26.1%	25.4%	19.9%	10.9%	53.6%	38.0%	35.1%	31.4%	59.5%	59.3%	57.5%	51.9%	100.0%	99.6%	83.8%	78.8%	62.0%	54.2%	46.2%	100.0%	100.0%	89.0%	86.8%	61.4%	61.4%	60.8%	99.8%	89.6%	65.0%	54.0%	28.2%	28.0%	25.8%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	GLA5 - conv - Ir0.002	-36.8%	51.1%	52.4%	37.5%	36.8%	33.4%	24.4%	20.3%	17.1%	28.1%	28.0%	19.6%	6.7%	45.7%	35.2%	32.7%	32.0%	63.6%	61.8%	60.0%	48.9%	100.0%	98.2%	67.0%	58.6%	37.8%	34.2%	28.6%	100.0%	100.0%	94.6%	85.0%	37.0%	38.6%	43.4%	99.2%	79.4%	42.4%	38.6%	21.0%	21.6%	24.8%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	GLA5 - Canon-AbCD - Ir0.005	-38.3%	55.8%	44.9%	38.2%	34.2%	38.2%	27.7%	23.3%	19.9%	25.9%	27.4%	20.3%	15.5%	51.9%	43.4%	37.4%	37.1%	53.5%	59.2%	56.5%	55.9%	100.0%	99.6%	98.0%	97.6%	91.8%	88.2%	85.6%	100.0%	99.8%	99.0%	93.6%	69.2%	60.6%	48.6%	100.0%	81.4%	18.0%	20.0%	17.0%	24.8%	23.8%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	GLA5 - Canon-AbCD - Ir0.003	-36.4%	52.0%	37.5%	27.9%	24.4%	32.6%	25.3%	20.0%	18.7%	24.5%	31.4%	26.0%	20.8%	56.4%	37.7%	32.0%	31.4%	62.8%	61.2%	56.5%	48.2%	100.0%	100.0%	98.2%	98.0%	92.0%	88.4%	75.6%	100.0%	100.0%	96.0%	95.2%	78.4%	76.2%	72.2%	100.0%	89.6%	30.6%	17.0%	23.2%	41.8%	39.4%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	GLA5 - Canon-ABCD - Ir0.002	-37.6%	43.1%	45.2%	41.5%	35.9%	35.1%	30.4%	22.6%	20.5%	25.3%	27.4%	20.6%	14.7%	49.5%	36.1%	34.6%	36.3%	59.3%	62.1%	59.1%	52.1%	100.0%	100.0%	91.8%	89.2%	69.2%	59.6%	47.8%	100.0%	100.0%	96.4%	94.8%	74.4%	49.2%	52.6%	99.8%	90.0%	69.0%	72.6%	49.2%	34.8%	30.6%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	GDN2 - conv - Ir0.005	-35.6%	48.2%	36.0%	32.6%	31.5%	35.2%	27.0%	20.4%	20.2%	24.6%	22.4%	19.3%	14.8%	49.0%	37.0%	36.2%	33.7%	59.5%	56.2%	54.2%	54.6%	100.0%	100.0%	98.4%	98.6%	95.4%	95.8%	96.0%	100.0%	100.0%	99.8%	99.2%	53.8%	54.6%	45.2%	100.0%	50.4%	10.0%	1.0%	1.6%	4.8%	3.2%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	GDN2 - conv - Ir0.003	-36.2%	47.8%	44.2%	34.6%	33.0%	37.8%	27.2%	23.1%	19.9%	25.4%	25.5%	22.0%	18.7%	50.8%	36.5%	30.9%	30.7%	52.1%	59.1%	54.7%	50.6%	100.0%	100.0%	99.6%	100.0%	98.2%	97.8%	96.0%	100.0%	100.0%	97.8%	83.8%	80.8%	32.4%	29.2%	99.4%	93.2%	67.6%	62.8%	49.2%	39.6%	34.8%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	GDN2 - conv - Ir0.002	-34.9%	50.1%	42.4%	33.9%	25.9%	36.6%	26.5%	18.1%	14.4%	25.2%	24.1%	19.6%	15.3%	54.1%	37.3%	37.9%	31.2%	47.7%	57.3%	53.6%	47.1%	100.0%	100.0%	100.0%	99.6%	99.6%	99.6%	100.0%	100.0%	100.0%	99.2%	98.0%	79.8%	72.2%	62.8%	99.8%	88.4%	66.0%	71.6%	41.8%	29.4%	23.8%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																														
1.3B	1.0T(16384)	GDN2 - Canon-AbCD - Ir0.005	-36.1%	43.3%	47.4%	39.3%	29.3%	34.0%	22.7%	21.2%	17.7%	27.9%	25.3%	21.3%	16.6%	51.9%	35.3%	37.1%	36.0%	66.5%	53.1%	51.4%	44.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	99.4%	95.2%	98.8%	69.2%	65.0%	100.0%	98.8%	94.4%	91.8%	77.4%	59.6%	64.0%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																													
1.3B	1.0T(16384)	GDN2 - Canon-ABCD - Ir0.003	-37.1%	48.6%	44.1%	36.7%	31.5%	30.3%	23.6%	20.5%	18.8%	29.4%	26.8%	25.7%	17.2%	54.4%	42.1%	38.3%	34.2%	55.6%	59.4%	55.1%	49.1%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	96.6%	99.0%	97.2%	90.8%	91.4%	100.0%	98.2%	18.2%	48.8%	12.8%	23.8%	40.4%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																													
1.3B	1.0T(16384)	GDN2 - Canon-AbCD - Ir0.002	-35.8%	45.4%	44.7%	39.0%	26.2%	38.8%	28.8%	20.1%	16.2%	24.2%	26.2%	23.1%	16.6%	45.9%	37.1%	35.2%	33.4%	64.1%	54.7%	49.4%	46.6%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	99.8%	100.0%	100.0%	97.0%	99.8%	80.6%	86.0%	77.8%	99.8%	98.4%	91.6%	90.0%	74.4%	79.8%	64.4%	0.0%	0.0%																																																																																																																																																																																																																																																																																																																																											
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